Mao Chen

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Research Interests

My research focuses on **structured representation learning** for visual and multimodal data, with an emphasis on structural reasoning across time, modalities, and viewpoints.

Education

MS **Institute of Automation, Chinese Academy of Sciences (CASIA)**, Pattern Recognition and Intelligent Systems — Beijing, China, Sept 2024 – June 2027
- **GPA: 3.97/4.00**

BS **Beihang University**, Computer Science and Technology — Beijing, China, Sept 2020 – June 2024
- **GPA: 3.86/4.00**

Publications

More Than Where You Are: Learning Semantics, Structure, and Geometry from Cross-View Localization — July 2026
M. Chen, X. Zhang, Z. Liu, C. Liu, X. Yang
arxiv.org/abs/2607.12429 (Under review for **IEEE TPAMI**)

Globally Localizing Lunar Rover in Pixels via Graph Alignment — Feb 2026
M. Chen, X. Yang, C. Liu, X. Zhang, X. Wang, Z. Bo, Z. Zhang, Z. Liu
arxiv.org/pdf/2606.10602

Fusion of Depth and Semantics for Probabilistic Floorplan Localization — Feb 2026
K. Ye, **M. Chen**, X. Zhang, X. Yang
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2026**)

Fine-Grained Multimodal Alignment for Image-Text Retrieval via Graph Learning — Jan 2026
M. Chen, X. Zhang, L. Qi, X. Li, X. Yang, S.C.H. Hoi, Z. Liu, M.H. Yang
10.1007/s11263-025-02686-y (International Journal of Computer Vision (**IJCV**))

Temporal-Aware Structure-Semantic-Coupled Graph Network for Traffic Forecasting — July 2024
M. Chen, L. Han, Y. Xu, T. Zhu, J. Wang, L. Sun
10.1016/j.inffus.2024.102339 (Information Fusion)

MFGCN: Multi-Faceted Spatial and Temporal-Specific Graph Convolutional Network for Traffic-Flow Forecasting — Dec 2024
J. Tian, L. Han, **M. Chen**, Y. Xu, Z. Chen, T. Zhu, L. Sun, W. Lv
10.1016/j.knosys.2024.112671 (Knowledge-Based Systems)

Sampling Spatial-Temporal Attention Network for Traffic Forecasting — Aug 2023
M. Chen, Y. Xu, L. Han, L. Sun
10.1007/978-3-031-40286-9_11 (International Conference on Knowledge Science, Engineering and Management)

Research Experience

Learning Semantics, Structure, and Geometry from Cross-View Localization — CASIA, 2025 – present
- Investigating **cross-view localization** as a learning framework for stable **semantics**, reliable **structure**, and transferable **geometry**, leading to a completed paper under review for **IEEE TPAMI**.

- Proposed **CROSS**, a unified framework with 3D-grounded alignment, structure-aware matching, and hypothesis ranking, **reducing localization error by 70%** at most compared to SOTA while achieving consistent cross-view understanding.
- Extended accurate localization to extreme lunar environments by proposing **Warped Alignment of Reprojected Graphs (WARG)**, which models rover scenes as reprojected graphs for view-invariant spatial matching.
- Achieved **1.68 m localization error** on the YuTu-2 dataset, significantly outperforming prior methods (~12.89 m), and led the lunar rover localization project as **first author**.

Fine-Grained Multimodal Alignment via Graph Learning — CASIA, 2024 – 2026

- Addressed limitations in fine-grained multimodal alignment, including **fragment-based fusion**, **redundant matching**, and **multimodal inconsistency**.
- Proposed a **graph-based alignment framework (GFGA)** to model correspondences between visual regions and textual components.
- Enabled fine-grained cross-modal interaction and disambiguation through **graph fusion and matching**.
- Achieved state-of-the-art results on MS-COCO and Flickr30K, with a first-author publication in **IJCV**.

Graph Neural Networks for Dynamic Spatiotemporal Forecasting — Beihang University, 2022 – 2024

- Investigated **dynamic spatial-temporal dependencies** in traffic systems, where static graph assumptions fail to capture evolving patterns.
- Developed **structure-semantic coupled** and **multi-faceted graph learning frameworks** for adaptive dependency modeling.
- Designed mechanisms for **dynamic graph construction** and **temporal interaction**, improving flexibility and robustness.
- Consistently improved prediction performance across benchmarks, contributing to multiple publications in **Information Fusion** and **Knowledge-Based Systems**.

Skills

Programming Languages: Proficient in Python, familiar with Java and C

Mathematical Foundation: Linear Algebra, Probability and Statistics, Calculus

Language: English (IELTS 7.5), Chinese (Native)

Awards & Honors

- Outstanding Graduate of Beijing (Top 5%) (2024)
- Academic Excellence Scholarship (2021-2023, Beihang University)
- Innovation and Entrepreneurship Scholarship (2023, Beihang University)
- Student Leadership and Service Scholarship (2021-2023, Beihang University)
- University Merit Student (2021 - 2023, Beihang University)

References

- **Prof. Xu Yang** | Email: xu.yang@ia.ac.cn | Affiliation: Institute of Automation, Chinese Academy of Sciences
- **Prof. Lu Qi** | Email: luqi360@whu.edu.cn | Affiliation: Wuhan University & Insta360